Embedded System HW Design
Week4
Types of Embedded Memory

- Identify different forms of Cache algorithms
- Discuss the advantages of having a MMU
- **Types of embedded memory**

Template designed by Tim Palmer tpalmer@vsu.edu

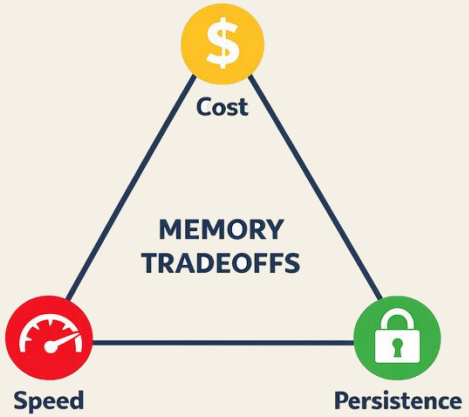
Embedded System HW Design
Week4
Why Memory Matters in Embedded Systems

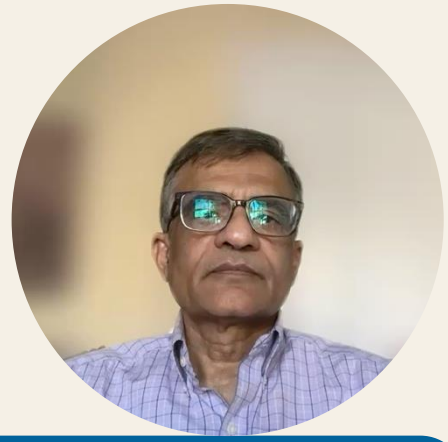
- **Memory affects**:
  - Performance and boot time
  - Power consumption
  - Cost
  - Reliability and data persistence

- **Common failure reason**:

  Not enough RAM or wrong memory type

- Good memory design balances speed, cost, and persistence





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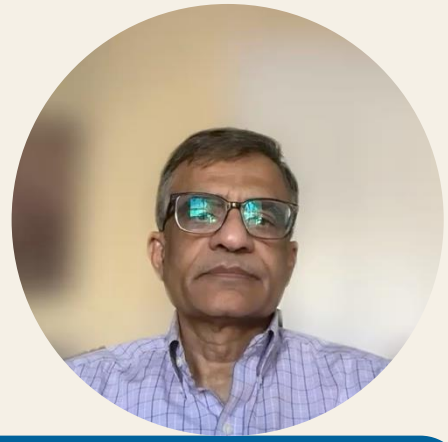
Embedded System HW Design
Week4
Memory Categories

Embedded memory is usually grouped into:

- **Volatile memory** (loses data when power is removed)

- **Non-volatile memory** (keeps data without power)



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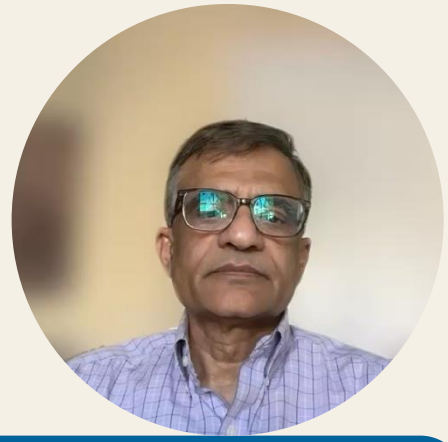
Embedded System HW Design
Week4
Flash Memory (Program Memory)

- **Non-volatile**: Stores firmware/program code
- **Used for**:
  - Application code
  - Constants
  - Bootloader
- **Tradeoffs**:
  - Slower writes
  - Limited write/erase cycles



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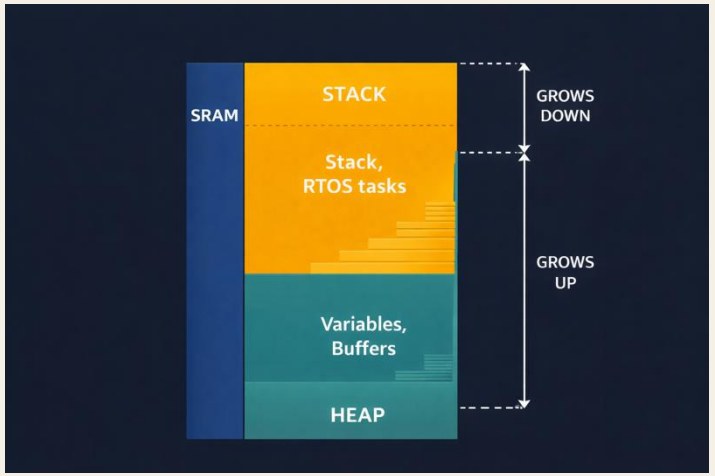
Embedded System HW Design
Week4
SRAM (Data Memory)

- **Volatile**: stores runtime data
- **Used for**:
  - Variables and Stack
  - Buffers (UART/SPI/I^2C)
  - Sensor processing data
- **Tradeoffs**:
  - Fast access
  - Limited size on MCUs



Stack + heap diagram

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Embedded System HW Design
Week4
EEPROM

- **Non-volatile:** Byte/Word write capability
- **Great for**:
  - Configuration settings
  - Calibration data
  - Device ID and small persistent data
- Slower writes than SRAM
- Often limited size (KBs)

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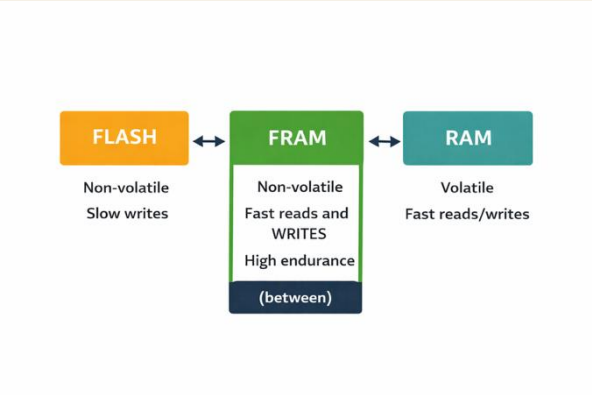
Embedded System HW Design
Week4
FRAM (Ferroelectric RAM)

- **Non-volatile** like Flash, but fast writes like RAM
- Very high endurance (many write cycles)
- **Great for**:
  - Frequent logging
  - Counters or event records
- More expensive than Flash
- FRAM is positioned between Flash and RAM



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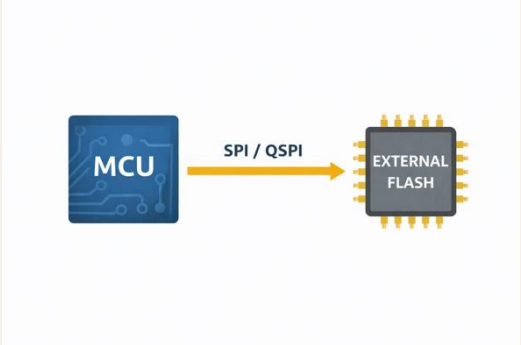
Embedded System HW Design
Week4
External Flash (SPI / QSPI Flash)

- **Non-volatile** like Flash, but fast writes like RAM
- **Non-volatil**e storage added externally
- **Used for**:
  - Large logs/data storage
  - Firmware updates
  - File systems
- **Interfaces**:
  - SPI (slower)
  - QSPI (faster)
- MCU → SPI/QSPI → external flash chip



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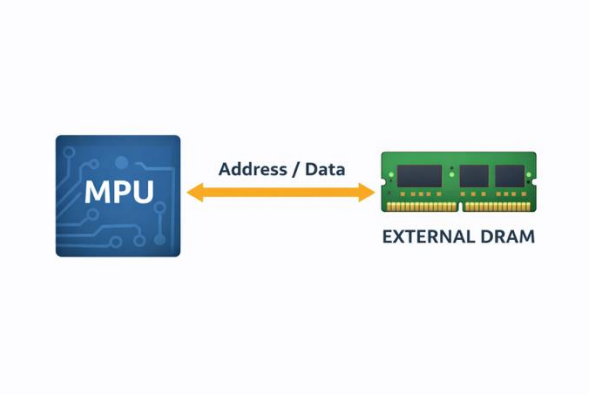
Embedded System HW Design
Week4
DRAM (Rare for MCUs, common for MPUs)
Volatile, high capacity

- **Volatile**: High capacity
- **Used when running**:
  - Linux
  - graphics/UI
  - large data buffering
- **Requires**:
  - Memory controller
  - More complex PCB routing
- MPU connected to external DRAM



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Embedded System HW Design
Week4
SD Card / eMMC Storage

- **Non-volatile**: Large removable or fixed storage
- **Used for**:
  - Data logging
  - File storage
  - Firmware storage
- **Tradeoffs**:
  - Slower access
  - Power spikes during writes
  - Wear-leveling concerns
- MCU → SDIO → SD card

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Embedded System HW Design
Week4
Memory Selection Tradeoffs

- Factors that drive memory choice:
- Size needed (KB vs MB vs GB)
- Speed (read/write)
- Power
- Endurance (write cycles)
- Cost
- Interface complexity

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Embedded System HW Design
Week4
Summary: What Memory is Used For

- **Flash** → Firmware/Program code
- **SRAM** → Runtime data and stack
- **EEPROM/FRAM** → Small persistent settings
- **External Flash/SD** → Large storage/logging
- **DRAM** → High-performance MPU systems

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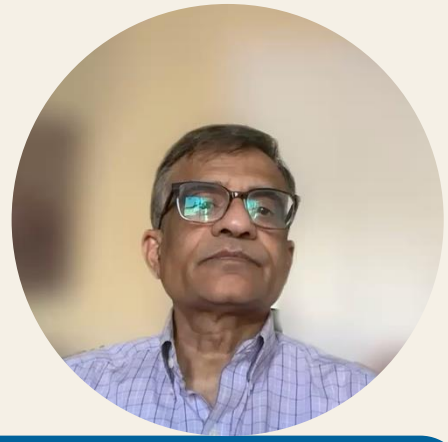
Embedded System HW Design
Week4
Common Embedded Memory Mistakes

- Underestimating RAM for buffers + RTOS
- Writing too frequently to Flash (wear-out risk)
- Ignoring power spikes during storage writes
- No plan for firmware update storage



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Embedded System HW Design
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